

TASK_TWO

2026-07-25

R Markdown

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When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(stringdist)
```

```
## Warning: package 'stringdist' was built under R version 4.5.2
```

```
library(tm)
```

```
## Warning: package 'tm' was built under R version 4.5.3
```

```
## Loading required package: NLP
```

```
## Warning: package 'NLP' was built under R version 4.5.3
```

```
library(proxy)
```

```
## Warning: package 'proxy' was built under R version 4.5.3
```

```
##  
## Attaching package: 'proxy'
```

```
## The following objects are masked from 'package:stats':  
##  
##   as.dist, dist
```

```
## The following object is masked from 'package:base':  
##  
##   as.matrix
```

```
#Load dataset  
amazon <- read.csv(  
  "C:/Users/MANASVI/Downloads/task2/Amazon.csv",  
  stringsAsFactors = FALSE,  
  fileEncoding = "latin1"  
)  
  
google <- read.csv(  
  "C:/Users/MANASVI/Downloads/task2/GoogleProducts.csv",  
  stringsAsFactors = FALSE,  
  fileEncoding = "latin1"  
)  
#Keep Required Columns  
amazon <- amazon[, c("id", "title")]  
google <- google[, c("id", "name")]  
  
#Clean the Product Names  
clean_text <- function(x) {  
  
  x <- iconv(x, from = "", to = "ASCII", sub = "")  
  
  x <- tolower(x)  
  
  x <- gsub("[^a-z0-9 ]", " ", x)  
  
  x <- gsub("\\s+", " ", x)  
  
  trimws(x)  
  
}  
  
amazon$clean <- sapply(amazon$title, clean_text)  
google$clean <- sapply(google$name, clean_text)  
  
#Test Fuzzy Matching  
stringsim(  
  "apple iphone 15 pro max",  
  "apple iphone promax 15",  
  method = "jw"  
)
```

```
## [1] 0.9173254
```

```
stringsim(  
  amazon$clean[1],  
  google$clean[1],  
  method = "jw"  
)
```

```
## [1] 0.4479974
```

```
#Build TF-IDF Matrix  
documents <- c(amazon$clean, google$clean)  
  
corpus <- Corpus(VectorSource(documents))  
  
dtm <- DocumentTermMatrix(  
  corpus,  
  control = list(  
    removePunctuation = TRUE,  
    removeNumbers = FALSE,  
    stopwords = TRUE  
  )  
)  
  
tfidf <- weightTfIdf(dtm)
```

```
## Warning in weightTfIdf(dtm): empty document(s): 411
```

```
tfidf_matrix <- as.matrix(tfidf)  
  
amazon_matrix <- tfidf_matrix[1:nrow(amazon), ]  
  
google_matrix <- tfidf_matrix[(nrow(amazon)+1):nrow(tfidf_matrix), ]  
  
#Cosine Similarity  
cosine_similarity <- function(x, y) {  
  
  if(sum(x) == 0 || sum(y) == 0)  
    return(0)  
  
  sum(x * y) / (sqrt(sum(x^2)) * sqrt(sum(y^2)))  
}  
cosine_similarity(  
  amazon_matrix[1, ],  
  google_matrix[1, ]  
)
```

```
## [1] 0
```

```
#Match Products
# -----
# Match Products (Optimized)
# -----

results <- data.frame()

# Cosine similarity matrix
cosine_matrix <- proxy::simil(
  amazon_matrix,
  google_matrix,
  method = "cosine"
)

cosine_matrix <- as.matrix(cosine_matrix)

for(i in 1:nrow(amazon)) {

  if(i %% 50 == 0){
    cat("Processed", i, "of", nrow(amazon), "\n")
  }

  # Top 20 TF-IDF candidates
  candidate_index <- order(
    cosine_matrix[i, ],
    decreasing = TRUE
  )[1:20]

  best_score <- 0
  best_match <- ""
  best_reason <- ""

  for(j in candidate_index){

    fuzzy <- stringsim(
      amazon$clean[i],
      google$clean[j],
      method = "jw"
    )

    tfidf_score <- cosine_matrix[i, j]

    confidence <- 0.5 * fuzzy +
      0.5 * tfidf_score

    if(confidence > best_score){

      best_score <- confidence
      best_match <- google$name[j]

      best_reason <- paste(
        "Fuzzy =", round(fuzzy,2),
        "| TF-IDF =", round(tfidf_score,2)
      )
    }
  }
}
```

```
}

results <- rbind(
  results,
  data.frame(
    Amazon_Product = amazon$title[i],
    Google_Product = best_match,
    Confidence = round(best_score,3),
    Reason = best_reason
  )
)
}
```

```
## Processed 50 of 1363
## Processed 100 of 1363
## Processed 150 of 1363
## Processed 200 of 1363
## Processed 250 of 1363
## Processed 300 of 1363
## Processed 350 of 1363
## Processed 400 of 1363
## Processed 450 of 1363
## Processed 500 of 1363
## Processed 550 of 1363
## Processed 600 of 1363
## Processed 650 of 1363
## Processed 700 of 1363
## Processed 750 of 1363
## Processed 800 of 1363
## Processed 850 of 1363
## Processed 900 of 1363
## Processed 950 of 1363
## Processed 1000 of 1363
## Processed 1050 of 1363
## Processed 1100 of 1363
## Processed 1150 of 1363
## Processed 1200 of 1363
## Processed 1250 of 1363
## Processed 1300 of 1363
## Processed 1350 of 1363
```

```
write.csv(
  results,
  "C:/Users/MANASVI/Downloads/task2/evaluation.csv",
  row.names = FALSE
)

file.exists("C:/Users/MANASVI/Downloads/task2/evaluation.csv")
```

```
## [1] TRUE
```